Soil Carbon (30 cm) Sampling
Supporting document `IndigoCarbon_US-1_2024_0001`

April 29, 2025

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1 Context

When soil sampling for Carbon by Indigo, all sampling locations require a soil carbon sample.

At each carbon sampling location, an individual 30 cm carbon sample must be collected according to the instructions outlined in this document. At a subset of carbon sampling locations, three different sample types are collected (figure 1) and the sample type bolded below must be collected according to the instructions in this document.

- **An individual 30 cm carbon sample according to the instructions below**
- An individual 30.48 cm bulk density sample according to the work instructions in `IndigoCarbon_US-1_2024_0002`
- A core that will contribute to a composite sample for pH and texture testing according to the work instructions in `IndigoCarbon_US-1_2024_0003` (applicable only to sites sampled March 16, 2020 – present)

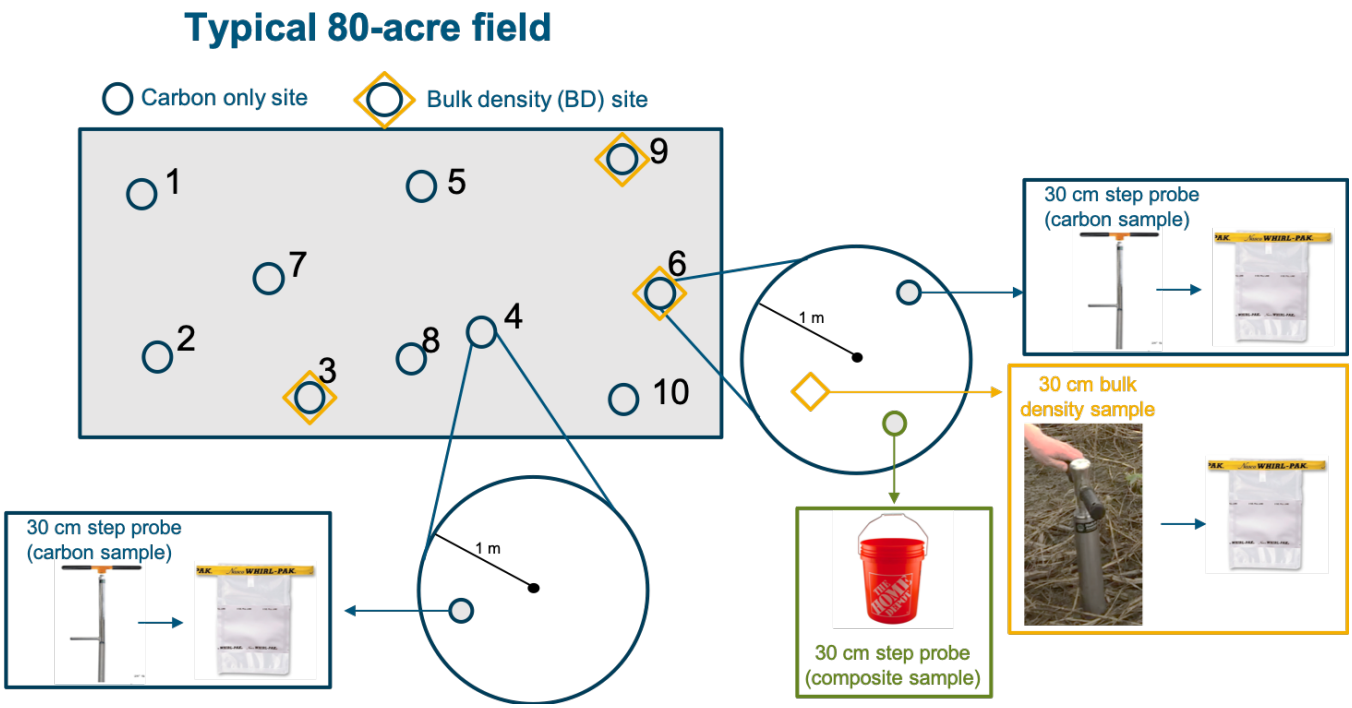


Figure 1: Indigo Carbon soil sample collection

(Representative of March 16, 2020 Current).

2 Inventory Checklist

	Tools	Quantity	Vendor	Catalog No.	Required/ Recommended
Probe Kit	33” plated step probe	1	AMS	401.40	Brand Recommended **Specifications Required:** • min. diameter: 7/8” • max. diameter: 1 1/4” • min. sampling section: 12” NOTE: diameter must be consistent across entire sampling section
	Soil extractor	1	AMS	402.02	Recommended
Drill kit	DeWalt brushless hammer drill	1	Home Depot	1001767317	Recommended
	DeWalt 20V Battery	3	Home Depot	202922414	Recommended
	DeWalt 20V Charger	2	Home Depot	205342899	Recommended
	Collect-N-Go Soil bucket	1	https://collectngonow.com	N/A	Recommended
	Collect-N-Go Soil 16” auger	2	https://collectngonow.com	N/A	Brand Recommended **Specifications Required:** • min. diameter: 7/8” • max. diameter: 1 1/4” • min. length: 16” NOTE: diameter must be consistent across entire sampling section of auger
	Rubbermaid 11.4 Qt tub	1	Amazon	FG2951-ARBISQU	Recommended
Sample Collection Kit	Whirl-pak bags 18 oz (blue)	1 box (500)	Nasco	B01341	Recommended
	QR stickers for Soil Carbon	Roll	Indigo	N/A	**Required**
	Soil Probe Brush	N/A	N/A	N/A	Recommended
	Putty Knife: 0.5”	N/A	N/A	N/A	Recommended
	Putty Knife: 1”	N/A	N/A	N/A	Recommended
	Canning Funnel	N/A	N/A	N/A	Recommended
	Rounded Wire Brush	N/A	N/A	N/A	Recommended
	Flathead Screwdriver	N/A	N/A	N/A	Recommended
	Sample Boxes	N/A	N/A	N/A	Recommended

Table 1: Checklist of tools and supplies used in bulk density sampling

3 Preparation

- Prepare sampling equipment before visiting the field. Review inventory checklist and make sure you have all necessary materials.
  - Take note of supplies in short inventory and request additional consumables from your Equipment Manager.
- All equipment can be stored in the trailer.
- Make sure you are familiar with all material in Indigo's field safety protocol. For this procedure you will need:
- NOTE: Safety rules and regulations of sampler's parent company should always supersede, when applicable, those recommended by Indigo.
  - Safety Glasses
  - Steel Toe Shoes
  - Cut-Resistant Gloves
- Navigate to the appropriate sampling location by following guidance in the IF app. When connected to the external GPS select the desired sampling point in the IF app and reduce the offset distance to that point a low as possible.
  - There are specific instructions in section 9 that will guide you when faced with situations when you are unable to navigate to the prescribed point due to any reason (flood, fence, private property) or find that the point is in a compromised sampling location (field edge, access road, rock outcropping).
- Clear residue or vegetation from the soil surface where you intend to take a soil core. Only sample bare mineral soil and avoid residue in the sample (figure 2).



Figure 2: Residue clearing to prepare sample core extraction

4 Assessment of Soil Conditions

- Excessively wet soils
  - Do not sample in standing water of any depth.
  - If upon removal of a soil core, the hole created is refilled within two minutes by surrounding soil flowing back into the hole because the soil exhibits a fluid-like property, discard the sample and discontinue sampling.
- Frozen soils
  - To avoid erroneous measurements of bulk density, we do not want to sample frozen soils.
  - When the daily average air temperature falls below 32°F, consult the instructions in section 8 to determine if samples can be taken.

4.1 Step probe sampling

- Using the step-probe, take a soil sample (one single probe) down to 30 cm depth from a random point within 1 foot (30.48 cm) of the plot center (figure 3)
  - You may wish to score or mark the probe with a 30cm mark to know the appropriate depth to take. **Do not rely on measuring the soil core itself for assessing adequate depth, as soil compaction during sampling will bias this measurement.**



Figure 3: Using a step probe to collect a 30cm soil sample

- Slowly remove the probe attempting to keep the sample intact and paying attention to any loss of soil or voids in the sample (figure 4)
- If you are unable to obtain a full 30 cm core due to rocks, wet soil, or loss of sample, attempt to take another sample within 1 foot (30.48 cm) of the original sample location.
  - If another attempt is required, do not go beyond 30cm of depth with the soil probe. Any sampling attempts to take a full core sample should only be conducted with the 0-30cm soil profile.
- After three failed sampling attempts, attempt to use the Drill Auger Sampling procedure described in section 4.2.
- Extrude the soil sample into a clean plastic bucket (2 gallon) (figure 5).
- Use the brush to clean any remaining sample in the probe into the bucket (figure 6).
- Pour through the funnel into the Whirl-Pak bag (figure 7).
- With the 1" putty knife, scrape the inside of the bucket clean of any sticking sample and add to the Whirl-Pak through the funnel again.
- Close the Whirl-Pak bag by flipping the bag twice and closing the tabs. Double-check that the bag is closed well and completely sealed shut to avoid spilling during shipping (figure 8).

4.2 Drill Auger Sampling

- In compacted, dry, or coarse textured soils, it may be difficult to sample using the step probe. In such cases, the drill auger approach may work better (figure 9).



Figure 4: Removing soil probe (left) and intact soil core (right)



Figure 5: Extruding soil sample into clean bucket

- Load the 5 ½" x 7/16" auger into the drill. Set the clutch setting to the highest level, set to drill mode so auto clutch does not engage and select forward drilling
- Mark soil auger with a Sharpie marker to indicate the relative position of the auger in relation to the bucket when a soil depth of 30 cm has been reached.
- Using the "Collect-N-Go" bucket kit, take a soil sample down to 30 cm depth from a random location within 1 foot (30.48 cm) of the plot center (Figure 11). Slowly auger with a downwards motion with repeated retrieval of the bit as you plunge to a depth of 30 cm (should be marked on the auger shank) while keeping the auger perpendicular to the ground. Be sure that you reach the correct depth.
- If needed, clean the auger of soil residue with the .5" putty knife into the bucket.
- Repeat the first step, 3 more times within a 1 foot (30.48 cm) radius of the original hole. (This should be enough material to fill the Whirl-Pak up to the white line).



Figure 6: Brushing remaining sample into the bucket



Figure 7: Pouring sample into a whirl-Pak bag

- ***If you encounter rocks that prevent you from reaching the full 30 cm on any of the repeated augering extractions, you must dump the entire bucket start the process again***
- Use the 0.5" putty knife to clean the auger after the last core has been collected
- After cleaning the auger, pour the sample through the funnel into the Whirl-Pak bag (figure 7)
- With the 1" putty knife, scrape the inside of the bucket to remove any residual sample and add to the Whirl-Pak through the funnel again.
- Close Whirl-Pak bag by flipping the bag twice and closing the tabs ensuring that the bag is completely sealed



Figure 8: Closing whirl-pak bag with sample

(figure 8)



Figure 9: Drill auger used to sample in compacted, dry, or coarse textured soils



Figure 10: Collect-N-Go bucket method for soil sample collection

5 Troubleshooting Carbon Sample Collection

What to do when a full depth sample is not possible:

- In particularly sandy, clayey, or rocky soils, collecting a full depth 30 cm carbon sample can be challenging; however, this measurement is crucial for our calculation of carbon stocks.
- As mentioned above, if a full depth carbon sample cannot be reached using the step probe, the sampler should attempt to apply the drill auger method.
- If on your first attempt to collect a carbon sample you are unsuccessful (assuming application of the most effective method):
  - Measure the depth of the collected sample and make a note of the length on the bag
  - Bag the sample and set aside
  - Attempt to collect a carbon sample to full depth two additional times at the prescribed location.
    - **NOTE: Always save the deepest sample obtained until a deeper or full depth sample is collected.**
- If a full depth sample cannot be collected at the prescribed location after three attempts:
  - Follow the offsetting protocol listed in the "Obstacles and Alternate Locations" section of Field Mapping and Navigation to attempt to locate a sampling point in which a full depth core can be extracted. One attempt should be made at each new offset location to obtain a sample at full depth until sampling of the point is completed.
  - **NOTE: Carbon samples, bulk density, and pH/texture samples must be collected at the same location. If the carbon sample is offset, the bulk density and pH/texture composite samples (when prescribed) must be obtained from the new offset location as well.**
- If a full depth sample cannot be achieved after offsetting:
  - Mark the QR code, sampling location, and depth of the deepest sample collected in the appropriate sections of the IF "Soil Sampling" form and proceed to the next prescribed sampling location.

        * **NOTE: Whenever possible, the core contribution to the texture/pH composite sample should be the same depth as the carbon sample. If faced with a scenario where this is not achievable, the deeper of the two cores should be designated as the carbon sample.**

6 Sample Labeling

- Make sure there is a QR code affixed to the bag
  - Note: if bag is wet, wipe off before attempting to add sticker, otherwise it may fall off during shipping
- With a sharpie write the following information as it appears in Indigo Fields Mobile (IFM) on the white section of the Whirl-Pak on the back of the bag (opposite from QR code label):

Minimum recommended labeling system:

- Sample Date (month/day)
- Field Name "Unique Full Word" (3 character minimum unless entire name is less)
- No multi-word abbreviations
  - From the following example:
    * For the field: Craig Place 1 - use "Place 1" **not** CP1
    * For the field: K Pivot SC - use "Pivot SC" **not** KPSC
- Sample Type (C, BD, pH/TXT) – **one code per bag**
- Sample Number
  - For carbon: C – 1,2,3,4...
  - For bulk density: BD – 1,2,3,4...
- **Include more info. if it helps you (e.g. grower name, farm name, full field name, etc.)**
  - **Minimum requirement example:**
    * For the first bulk density sample collected on the field Crabbe C6C on 5/13:
      · 5/13 - C6C – BD1 (figure 11)
  - **More info. example:**
    * For a carbon sample collected at the fifth spot label as follows. **Grower**: Fred **Farm**: McDonalds **Field**: Home **Sample:** C5
      · Fred-McDonalds-Home-C5

6.1 Sample Entry into Indigo Fields

- Use Indigo Fields (figure 12) to find the task associated with the field you are sampling, and select the correct survey for the sampling activity
- Use an external GPS device (Bad Elf, Garmin, or similar) that is connected via Bluetooth to your iPad.
- Select the point you want to enter data on, (figure 12) answer the field-level Survey Completion questions.
- Proceed through the general questions (field-level). (figure 13)
  - For each point, there are general questions you must answer ex. are you able to take this sample? as well as a location verification. Click "Record Location" once you are near the point - the app will save your location and tell you how close to the point you were when you clicked "record".
    * Get as close to the prescribed point as possible - the app will record your location as well as the location that was prescribed.

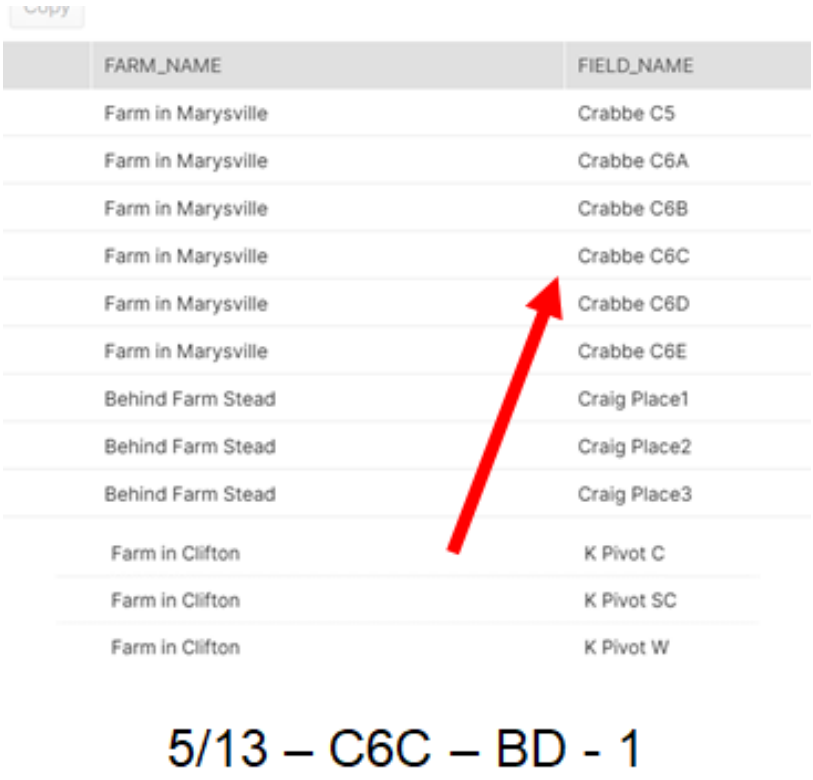


Figure 11: Labeling the bulk density sample

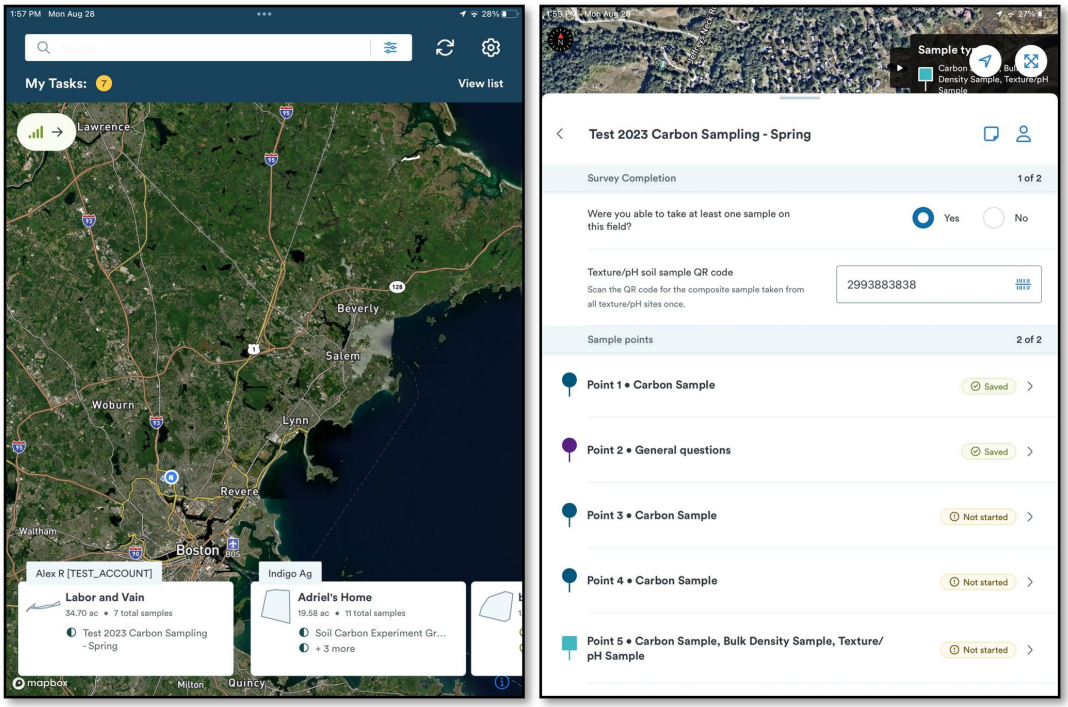


Figure 12: Using IF to record a new soil sampling event, a, left shows regional map view, b, on right displays samples at a location

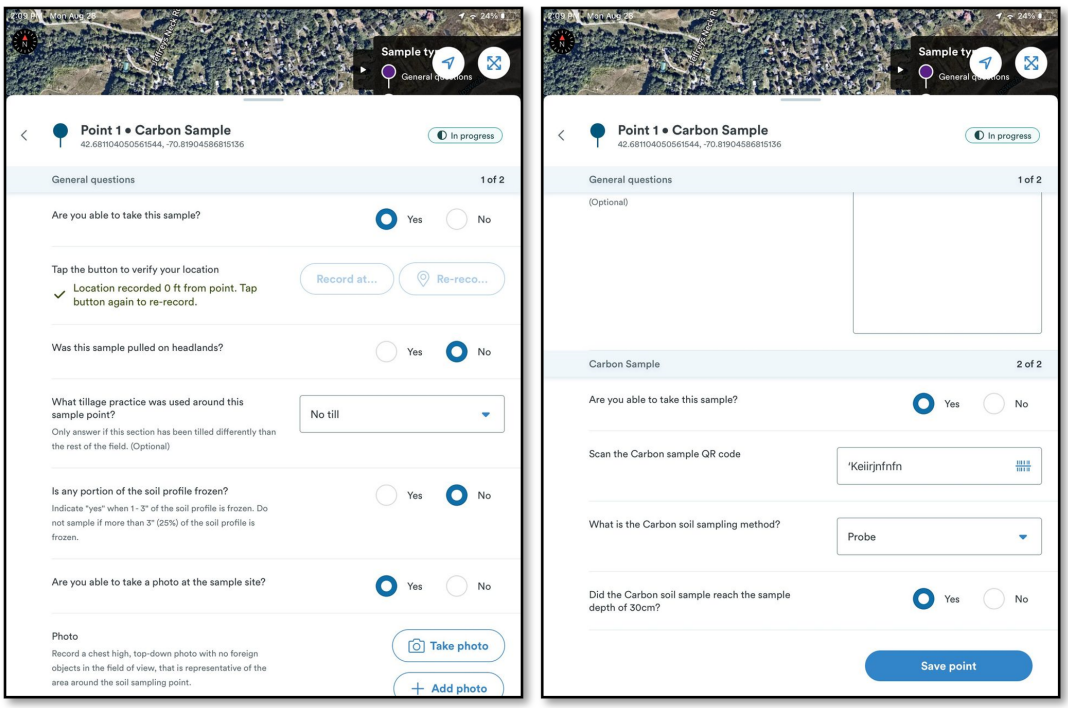


Figure 13: Recording Sample Parameters in Indigo Fields 2.0

  - ∗ If you are unable to sample at the prescribed location, choose a reason why the sample was offset from the drop-down menu. If you are $>$ 20 feet from the prescribed location, the app will force you to choose an option. If you are $<$ 20 feet, you can choose to enter an offset reason.

- Answer questions about location of the sample and whether the soil was frozen or not. If more than 3" of the soil profile is frozen, do not sample - mark a reason why you cannot sample the field or point. (figure 13)
- Answer the optional tillage practice question only if tillage is inconsistent across the field (see section 7 with more details.
  - If tillage is inconsistent across the field (eg: some portions of the field appear to be no-till while others appear to have been conventionally tilled) this information should be recorded at the point level for all samples.
  - If tillage is the same across the field, this question does not need to be answered.
- Take a photo of the sample site.
- Add any relevant notes.
- Proceed through the Carbon sample questions. (figure 13)
  - Answer if you are able to take the sample or not (if not, choose a reason why not)
  - Scan the QR code of the soil sample bag from this point.
  - Select the soil sampling method.
  - If the full depth was achieved (30 cm) select "yes" to "Did the Carbon soil sample reach the sample depth of 30 cm?"; if not, select "no", enter the depth reached, and select a reason why the full depth was not reached. (figure 13)
- Save the point, and you will return to the list of points to be sampled on the field.
- Proceed to enter data at each prescribed point around the field. (figure 12)

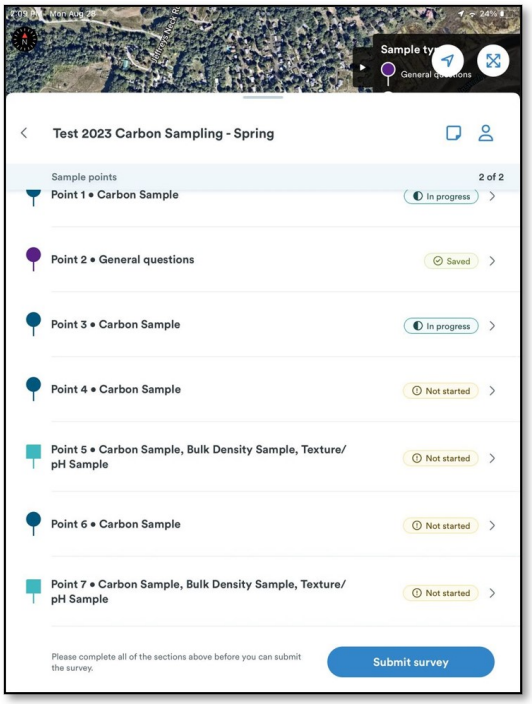


Figure 14: Status view during location sampling

- When you are done (all points should show SAVED), make sure to "Submit Survey" at the bottom of the screen (figure 14).

7 Inconsistent Tillage Instructions

- If tillage is inconsistent across the field (e.g. some portions of the field appear to be no-till while others appear to have been conventionally tilled) this information should be recorded at:
  - **Point Level** (May 12, 2020 – September 12, 2020)
  - **Bulk Density Sample Sites Only** (September 13, 2020 – Present)
- Scroll to the bottom of the Soil Sampling form to the Point Level Crop/Soil Information section and fill in the type of tillage used at each individual sampling point using the dropdown menu (Figure 13)
- As a general guideline see figure 15:
  - Conventional tillage will look like there have been many passes with a plow and there is very little crop residue remaining on the field.
  - Minimal tillage will look like there has been a single plow pass on the field, and some (~30%) crop residue may remain
  - Strip tillage will have distinct strips of tilled areas and undisturbed areas
    - **Please remember that if the field has been strip tilled, bulk density samples should be pulled from within the undisturbed strips.**
    - Strip tillage across the entire field does not require point level tillage information
  - No-till will show no evidence of disturbance and near complete coverage with crop residue



Figure 15: Typical disturbance and residue left on field with various tillage methods

8 Frozen Ground Procedures

8.1 Suggested FOS dispatching guidelines

- **When daily average air temperatures fall below 32°F (0°C) the FOS Dispatcher should progressively consult the parameters below**, based on the availability of information, to determine if a FOS should be dispatched to further assess soil conditions.
- Soil Temperature Resources
  - Use Syngenta GreenCast to determine the soil temperature of the top 0 - 4 inches (10.16 cm) of soil. **If soil temperature is less than 28°F (~-2°C), do not dispatch**
  - http://www.greencastonline.com/tools/soil-temperature
- Air Temperature Resources
  - Use only if soil temperature information is not available
  - **If the air temperature has been below 28°F (~-2°C) consecutively for 7 days or more, do not dispatch**
  - National Weather Service: https://www.weather.gov/
- If none of the above criteria are met, sampling may be assigned and the FOS should follow the procedure outlined in the following section. Exercising good judgement in regards to FOS personal safety is always strongly advised.

8.2 In field assessment of frozen ground

If the daily air temperature falls below 32°F, FOS should assess whether the ground is frozen before proceeding with sampling

Soil Surface Inspection (October 1, 2019 – August 31, 2020)

1. Use the auger bit to probe the soil surface.
2. If the surface layer of the soil is frozen to any depth (Figure 16), record Yes in the appropriate IF 2.0 field (Figure 13) within the Soil Sampling call plan asking "Is the surface of the soil frozen?"

Soil Profile Inspection (October 1, 2019 – Present)

1. Attempt to collect a 30 cm soil core using the step probe. If successful, proceed to 3. If unsuccessful proceed to 2.
2. If using the step probe proves unworkable, utilize the auger method outlined in the "Compacted, Dry, or Coarse Textured Soils Unable to Be Sampled by Soil Step Probe" section.
3. Visually and physically inspect the profile to determine the extent of soil freezing (Figure 16). **Sampling should not be conducted if more than a total of 25% (~7.5 cm) of any portion of the 30 cm soil profile is found to be frozen or containing frost.**

- If less than 1" (2.5 cm or ~8%) of the soil profile is frozen (e.g. frost at the soil surface). Proceed as if the soil at the sample point is not frozen.
- If greater than or equal to 1" (2.5 cm or ~8%) but less than or equal to 3" (7.5 cm or ~25%) of the total soil profile is frozen, mark yes in the IF field Is any portion of the soil profile frozen and proceed with sampling using the sampling method best suited to soil conditions.
- **This information should be recorded at the sampling point level.** Within the Soil Sampling form, scroll down to the Point Level question: "Is any portion of the soil profile frozen?" (Figure 13).



Figure 16: Frozen Soil

9 Obstacles and Alternate Locations

October 1, 2019 – February 28, 2021

1. Selecting an alternative location when an obstacle is encountered:
    1. If the obstacle is a permanent road, pond or other large, non-productive area of the field, the field boundary should be adjusted according to the directions outlined above in "Navigating to Field."
    2. When you encounter a smaller, localized obstacle such as a large stone or rock pile, a tree, a fence line, small wet area, or other immovable obstacle in an area that should remain part of the productive field or pasture, you will need to select an alternative location at random as outlined below.
    3. To select the alternative location, travel due north 25 feet using a compass application on the iPad. If you reach the field boundary, turn 15 degrees clockwise and repeat. If you have still not cleared the obstacle repeat the procedure at 50 feet (Figure 6).

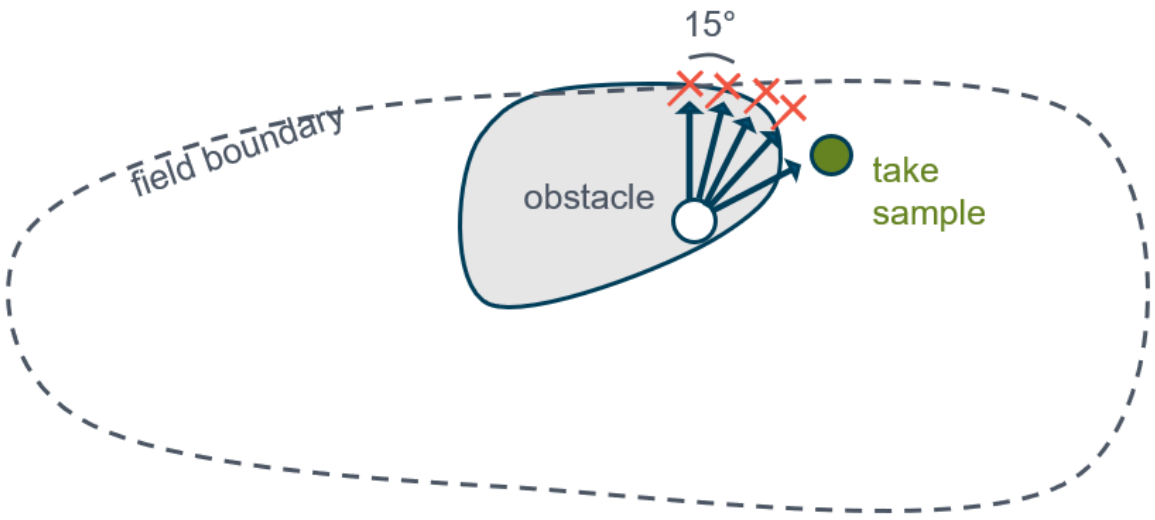


Figure 6. Sample offsetting procedure (October 1, 2019 – February 28, 2021)

2. After completing the sampling procedure, record a sampling event in Indigo fields mobile using the following steps:
    1. Select the point location you are sampling.
    2. Navigate to the alternate location using the procedure above.
    3. Choose the "Record offset location" option in IF. (This will connect the intended location with your actual location.
    4. Scan the barcode of the sample/samples you just extracted.
    5. Enter the other required information in the soil sampling event form as indicated in `IndigoCarbon_US-1_2024_0001` and `IndigoCarbon_US-1_2024_0002`
    6. Record reason for off-setting using the 'reason for not retrieving soil sample' drop-down menu (Applies only to samples collected September 1, 2020 – February 28, 2021)
    7. Save the sampling event.
    8. If you are unable to collect a sample using the offsetting procedure, follow the guidelines outlined in "Documenting Prescribed Sampling Points Unable to be Collected" below (Applies only to samples collected April 29, 2020 – February 28, 2021).

March 1, 2021 - Present

1. Selecting an alternative location when an obstacle is encountered or a sample cannot be collected to full depth:
    a. If the obstacle is a permanent road, pond or other large, non-productive area of the field, the field boundary should be adjusted.
    b. When you encounter a smaller, localized obstacle such as a large stone or rock pile, a tree, a fence line, small wet area, other immovable obstacle or a sample cannot be collected to full depth in an area that should remain part of the productive field, you will need to select an alternative location at random by proceeding to #2.
2. To select the alternative location, travel due north 25 feet from the prescribed location using a compass application on the iPad. If you reach the field boundary, fail to clear the obstacle, or still cannot collect a sample to full depth, turn 90 degrees clockwise and repeat (Figure 7).

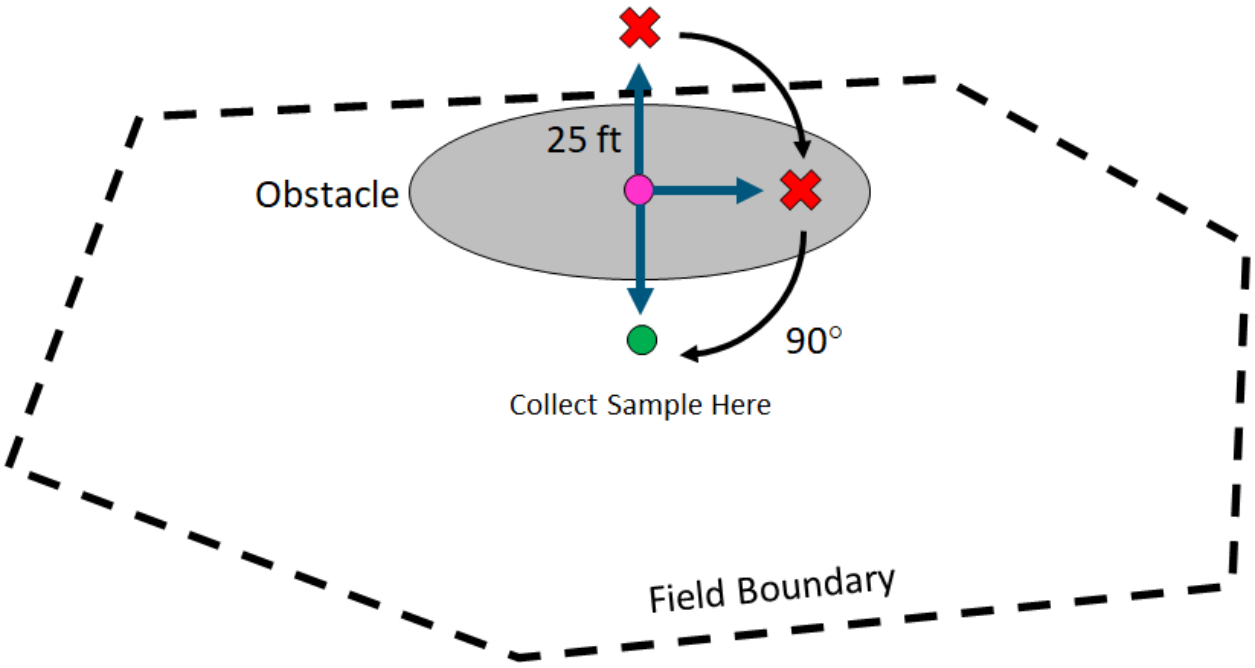


Figure 7. Sample offsetting procedure (March 1, 2021 – Present)

3. After completing the sampling procedure, record a sampling event in Indigo Fields using the following steps:
   a. Select the prescribed point location you are sampling.
   b. Stand at the alternate location selected using the procedure above.
   c. Choose the "Record offset location" option in IF to connect the intended location with your actual location.
   d. Scan the barcode of the sample/samples you just extracted.
   e. Enter the other required information in the soil sampling event form as indicated in `IndigoCarbon_US-1_2024_0001` and `IndigoCarbon_US-1_2024_0002`
   f. Record reason for off-setting using the 'reason for offsetting soil sample' drop-down menu
   g. Save the sampling form.
4. If you make a complete 360° revolution around the prescribed point and have still not cleared the obstacle or collected a sample to full depth:
   a. If a sample cannot be collected at all:
      i. Follow the guidelines outlined in "Documenting Prescribed Sampling Points Unable to be Collected"
   b. If a sample can be collected, but not to full depth:
      i. Retain and record the deepest sample achieved as indicated in `IndigoCarbon_US-1_2024_0001` and `IndigoCarbon_US-1_2024_0002`. If the sample was collected in an offset location, follow the instructions outlined in #2 above.
      ii. **NOTE: Whenever possible, the core contribution to the texture/pH composite sample should be the same depth as the carbon sample. If faced with a scenario where this is not achievable, the deeper of the two cores should be designated as the carbon sample.**

10 Documenting Prescribed Sampling Points Unable to be Collected (April 29, 2020 – Present)

If you are unable to collect a sample, follow the guidelines outlined below:

1. Sample as many points as possible under the current SOP guidelines
2. Enter "ERROR" (all letter capitalized) into the QR code entry field of the Indigo Fields call plan
3. Select the appropriate explanation from the "Reason for not retrieving soil sample" dropdown menu according to the following:

- **Established crop** – Sample point is in an area of active crop growth and is inaccessible.
  - When possible, return to the field to collect this sample once accessible.
  - When possible, return to the field to collect this sample.
- **Livestock in area temporarily** – Livestock are currently occupying the sampling area. FOS access is restricted.
  - When possible, return to the field to collect this sample once accessible.
- **Permanently un-sampleable and unable to offset** – This designation is only appropriate when the offsetting protocols absolutely cannot be executed successfully
- **Point outside of boundary** – Sample point is located outside of the specified field boundary.
  - Report to Indigo Fields support team.
- **Seasonal delay** – A major event, such as flooding, has rendered the sample point inaccessible for the remainder of the sampling season. The point likely cannot be sampled within the same spring or fall sampling campaign as the rest of the field.
  - When possible, return to the field to collect this sample once accessible.
- **Short-term weather delay** – Sample point is not able to be collected along with the remaining points on the day of sampling but likely will become available within 14 calendar days.
  - When possible, return to the field to collect this sample once accessible.
- **Unable to sample to full depth and depth was not adequate** - Methods used to relocate point failed.
  - Report to Indigo Fields support team
- **Point not in planting area** - Point falls in an area of the field not actively cropped.
  - Relocate point using procedure described earlier in this document, if boundary needs to be edited, report to Indigo Fields support team.
- **Soil is frozen** - Indicates that soil was frozen to a level greater than permissible in this protocol.
  - Return to sample when conditions permit.
- **Soil too compacted, multiple tries attempted** - Attempts to get a full depth sample using point relocation methods described earlier in this document failed due to extreme compaction.
  - Report to Indigo Fields support team.
- **Field was being managed (combine, chemical application, irrigation, etc.**
  - Plan to return when conditions permit and are safe.
- **Field inaccessible (locked gate, road too muddy)**
  - Return to sample later if possible or report to Indigo Fields support team.
- **Equipment Failure**
  - Return to sample once equipment can be repaired.
- **Standing Water** - moisture levels exceed acceptable levels described in this document.
  - Return to sample once field has dried out.

11 Summary of Changes

Section	Date	Summary of Changes
Throughout	May 12th 2024	Refrences and figures updated to reflect app changed from Indigo Fields Mobile to Indigo Fields 2.0 (IF 2.0).